

The big picture

Guiding question(s)

- What determines the metallic nature and properties of an element?

Keep the guiding question in mind as you learn the science in this subtopic. You will be ready to answer it at the end of this subtopic. The guiding question requires you to pull together your knowledge and skills from different sections, to see the bigger picture and to build your conceptual understanding.

Look around your surroundings and see whether you can identify any metal objects. Whether looking inside your home or outside in a big city, metal objects can be commonly found as appliances, building structures, vehicles, electrical wiring and jewellery. How can metals create such a wide range of objects with varying sizes, shapes and functions?

Gold is a metal that is highly valued and very rare to find. Many have encountered gold in ornamental objects, jewellery and investments. Gold is extraterrestrial, meaning that it exists on Earth only because meteorites containing gold bombarded the Earth's surface about 4 billion years ago. This bombardment embedded gold into the earth's mantle and crust. **Figure 1** shows rocks and ores with gold deposits resulting from this meteorite bombardment.



Figure 1. Ore containing gold.

Credit: Photography by Mangiwau, [Getty Images](https://www.gettyimages.co.uk/license/911604090)

[.https://www.gettyimages.co.uk/license/911604090](https://www.gettyimages.co.uk/license/911604090)

Gold mining is the process of physically removing rocks containing gold deposits from the earth. When it is noticed that the surface of the Earth's crust has gold embedded in it, a mine is created, where explosives are used to fragment the rock so that it can be easily transported. **Figure 2** shows an example of an open pit gold mine. To extract the gold from the rock, the rocks are first ground down into a fine powder, filtered and heated above gold's melting temperature where molten gold can be poured into moulds.

The process of mining gold has been used for thousands of years. The scarcity of gold has made it a valuable commodity that has global economic benefit. However, the environmental effects of mining can include erosion, loss of biodiversity and deforestation. A global tension exists between mining for economic benefit and negatively affecting our environment.



Figure 2. An open-pit gold mine.

Credit: John W Banagan, Getty Images

(<https://www.gettyimages.co.uk/detail/photo/singel-penny-with-water-droplet-royalty-free-image/133896288>)

It is important for us to consider that every metallic object that we regularly interact with has been mined from the Earth. These metals provide useful materials, but it comes at the cost of our natural resources. Reducing our need for overconsumption and recycling are ways that we can individually take action to reduce the demand on the Earth's resources.

As we explore the structure and bonding of metals, consider what questions you might have. Why are they shiny? Why do we use them for electrical wiring? If you are outside on a hot day, why does a metal chair feel so much hotter than its surroundings? Why do we use metals for cooking and eating? Why are car bodies made of metal? Keep any questions you have fresh in your mind as we explore metallic bonding!

Prior learning

Before going further make sure that you understand the following:

- Be able to identify the region of the periodic table where metals are found (see [section S2.1.1](#) <https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43111/print/>)

[426-cid-753301/book/formation-of-ions-id-43120/](#))).

- Be able to deduce the number of valence electrons and electron configuration for elements based on their position on the periodic table (see [section S1.3.5](#) [!\[\]\(633dd45d48d71eb51a85c6dd83ee51e9_img.jpg\) \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electron-configurations-id-43569/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electron-configurations-id-43569/))).
- Identify metals as elements that typically lose electrons (see [section S2.1.1](#) [!\[\]\(bdddf9191a284aa0945448444083c5b0_img.jpg\) \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/))).